

MICHAEL LUSTIG

Senior Software Engineer | Remote/EST

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SUMMARY

Communication-minded senior full-stack and mobile engineer across React, React Native, iOS, and Android, specializing in AI-native, agent-addressable product systems. Uses clean architecture to make applications predictable, observable, understandable, and replayable both by humans and LLM agents alike. Crafts architectural solutions that let LLM agents and humans run core business logic outside heavy UI frameworks through CLI-enabled workflows. This approach both rapidly improves iteration and agility and provides a much more enjoyable developer experience. These strategies prevent duplication and drift when new (especially LLM-driven) features are retrofitted onto existing products. Solves problems methodically and turns architectural decisions into systems teams can inspect, debug, and extend.

SKILLS

AI/Product: AI-native architecture; AI addressability; LLM integration; LLM observability; LLM tracing; local agent tooling; skill creation/maintenance; agent-compatible debugging; ADR automation; cross-team product delivery.

Stack: React Native; React; TypeScript; Swift/SwiftUI; TCA; SPM; Python; GraphQL; gRPC/Protobuf; REST/Express; XState; React Query; Expo; Zustand; CLI tooling; dependency isolation.

Web3/Crypto: Solana; Ethereum; Solana programs; Solana Pay; Anchor; smart contracts; NFTs; Privy embedded wallets; XMTP; thirdweb; passkeys; account abstraction; gas sponsorship; non-custodial wallet linking.

Other Tools: InstantDB; Firebase; Firebase App Check; Langfuse; OpenTelemetry; Deepgram; OpenAI APIs; Expo Native Modules; App Attest/DeviceCheck/Play Integrity; React Native WebView; WebSockets; Docker; ngrok; Bun; Jest; Zod; Unleash; GitHub Actions.

EXPERIENCE

Shopify

March 2026 - July 2026

Senior Software Engineer - React Native

- Prototyped AI-native mobile architecture patterns that made app state and actions addressable by LLM agents across login, home, product-list, and product-detail flows.
- Built local Sidekick developer tooling so a development agent could exercise in-app AI behavior through the mobile WebView via CLI, improving mobile debugging and iteration cycles.
- Led mobile product-collection features through incomplete backend APIs, web/mobile versioning differences, and GraphQL/header/payload debugging; built agent-compatible CLI utilities to probe endpoints with the same authentication, headers, versions, and payloads used by the mobile app, further improving iteration and backend debugging time.
- Modularized identity/authentication, product-list, and product-detail logic so core behavior could run outside the full mobile binary in local, CLI, and CI contexts.

Oscar Health

April 2025 - March 2026

Senior Software Engineer

- Designed healthcare member lookup workflows using acronym expansion, fuzzy matching, LLM synthesis, caching, cross-tab synchronization, and custom React hooks; improved call resolution by roughly 12% in A/B tests and reduced average call time by roughly 2.5 minutes from an approximately 38-minute baseline.
- Led Follow-Up Actions Architecture & Implementation for an LLM-driven member interaction system across LLM agent system, Python, gRPC, React, and React Native; mapped protobuf/gRPC responses into shared React and React Native view models through a pure JavaScript library.
- Expanded an LLM-assisted review moderation workflow from binary terms-violation labels into rule-level reasoning and feedback-auditing loops; reached 100% agreement on the human-evaluated set and surfaced previously manually approved edge cases for review.
- Collaborated across Developer Operations & Infrastructure (Kubernetes, Terraform, bespoke YAML configuration) to set up self-hosted Langfuse for LLM observability, tracing, cost tracking, analytics, and feedback collection, sidestepping the need to use a third-party SaaS dependency for sensitive healthcare-adjacent workflows.

Ephemera

September 2024 - February 2025

Staff Frontend Engineer

- Encapsulated XMTP SDK consumption behind a dependency layer and dedicated client manager, modeling

active-account invariants at the right boundary and exposing protocol-shaped dependencies so business logic was easier to reason about, mock, test, and run independently of the React Native binary.

- Extracted Privy embedded-wallet creation, Web3 identity, account abstraction, gas sponsorship, Ethereum/XMTP account creation, later Solana support, and cross-device linking into XState machines with dependency injection.
- Built React Native/Expo architecture for Convos, including Ethereum address search, optimistic mutations, cache reads, data normalization, thin-client boundaries, thirdweb/non-custodial wallet linking, and Firebase App Check token generation.
- Championed ADRs and LLM-assisted transcript workflows that turned architecture discussions into agent-readable decision records with documented tradeoffs, North Stars, and maintenance context.

Jabra Hearing

August 2023 - September 2024

Senior React Native Engineer

- Diagnosed Bluetooth pairing loops between parent and child XState machines by synchronizing state-machine data to a WebSocket visualization and combining it with log inspection; refined state charts to improve hearing-aid pairing robustness.
- Built a pure-JavaScript, Zustand-like state management library with Zod validation, persistence, schema migrations, observable stores, and fakeable dependencies.
- Completed the XState v4-to-v5 migration across approximately 25 to 30 machines by prototyping on critical pairing logic, running v4 and v5 side by side, migrating tests, and monitoring analytics for regression signals, without reported production regressions.
- Replaced leaky React-dependent static classes with schema-backed observable stores, making side effects explicit and simplifying tests, demo mode, and non-React execution paths.

Bakkt

August 2020 - March 2021

Full-Stack Software Engineer

- Led Bakkt Wallet and Application Gateway Proxy work with React Native, Node.js, Apollo GraphQL, and finite state machines for cryptocurrency transaction and bank-linking flows.
- Built native Objective-C/Java React Native modules for Google Wallet and Apple Pay integrations.
- Partnered with design leadership on scalable DesignOps workflows and built foundational shared component libraries.

OTHER EXPERIENCE

Prosper Marketplace: Staff Frontend Engineer, March 2022 - August 2023.

TCT Ministries: Android, iOS, React Native Developer, January 2015 - July 2020; cross-platform media catalog and streaming applications across mobile and backend systems.

SELECTED CRYPTO PROJECTS

Proof of Listening: Lead Full-Stack / Smart Contract Developer, Solana Grizzlython, 2023; built a full-stack audio recognition pipeline for validating that a user actually listened to a piece of audio, with a custom Solana program/smart contract, NFT minting, dynamic on-chain pricing, React Native, Rust, TypeScript, Firebase, ShazamKit, and whisper.cpp.

mtnBooth Twitter Solana Pay Integration: Lead Developer, 2022; built a Next.js/Open Graph app for minting mtnFrens NFTs with Twitter integration, Solana, and Solana Pay.

SELECTED AI TOOLING

Organic Software: SwiftUI/TCA iOS app for real-time speech transcription, LLM processing, code modifications, and PR generation with OpenAI APIs, WebSockets, and Apple Speech Framework.

GPT Custom Server: Early local AI tooling that enabled ChatGPT-style workflows to interact with a local filesystem and run tests, implemented with TypeScript, Express, Docker, ngrok, Bun, Jest, and Bash.

EDUCATION

Florida State University: B.S. in Computer Science, 2011 - 2015; minor in Mathematics and Physics.